

Scheduling in Batch Systems

First-come first-served, shortest job first, and shortest remaining time next

Definition

Batch scheduling chooses the order in which noninteractive jobs run. Since users are usually not waiting for immediate responses, batch schedulers often focus on throughput, turnaround time, and CPU utilization.

1. Where Batch Scheduling Fits

Main Idea

Batch systems run jobs submitted for later completion, such as payroll, inventory, banking calculations, or insurance claim processing.

Different From Interactive Scheduling

In batch systems, there is usually no user sitting at a terminal expecting a quick response to each command. This makes nonpreemptive algorithms or long time slices more acceptable.

Some scheduling algorithms are useful in both batch and interactive systems, but the classic batch algorithms are easiest to understand first.

2. Batch Scheduling Goals

Goal	Meaning
Throughput	Complete as many jobs per hour as possible.
Turnaround time	Minimize the time from job submission to job completion.
CPU utilization	Keep the CPU busy while avoiding excessive switching overhead.

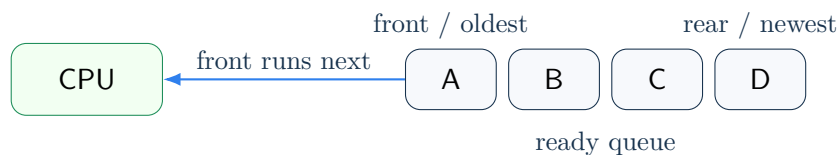
3. First-Come, First-Served

FCFS

First-come, first-served, or **FCFS**, is a nonpreemptive scheduling algorithm that gives the CPU to processes in the order they request it.

Ready Queue

FCFS uses a single queue of ready processes. New jobs and newly unblocked processes are added to the end of the queue. The scheduler removes the next process from the front.



Nonpreemptive Rule

Once a process starts running under FCFS, it is allowed to run until it blocks or finishes. It is not interrupted just because it has run for a long time.

4. FCFS Advantages

Advantage	Explanation
Simple	The scheduler only needs a FIFO queue.
Easy to implement	Add to the back, remove from the front.
Fair in arrival order	Jobs are served in the order they requested the CPU.
Low overhead	It avoids frequent preemptions and context switches.

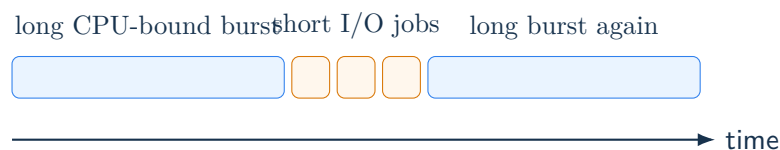
5. FCFS Disadvantage

Convoy Effect

FCFS can make many short I/O-bound processes wait behind one long CPU-bound process. This is often called the **convoy effect**.

Why I/O-Bound Jobs Suffer

Suppose one CPU-bound process runs for 1 second at a time, while many I/O-bound processes need only a little CPU time before starting disk reads. Under FCFS, the I/O-bound processes may wait behind the long CPU burst again and again.



Better With Preemption

If the long CPU-bound process were preempted every 10 msec, the I/O-bound processes could issue disk requests much sooner and finish far faster.

6. Shortest Job First

SJF

Shortest job first, or **SJF**, is a nonpreemptive batch scheduling algorithm that runs the waiting job with the shortest known running time.

Main Assumption

SJF assumes the scheduler knows the run time of each job in advance. This can be realistic in repeated batch workloads where similar jobs are run regularly.

Insurance Company Example

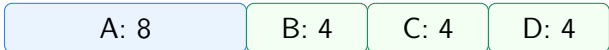
If an insurance company runs the same type of claim-processing batch every day, it may be able to estimate how long that job will take.

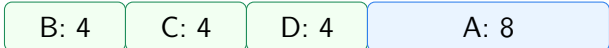
7. SJF Turnaround Example

Original Job Lengths

Suppose four jobs have running times:

$$A = 8, \quad B = 4, \quad C = 4, \quad D = 4$$

(a) Original 

(b) SJF 

—————→ time

Order	Completion Times	Average Turnaround
A, B, C, D	8, 12, 16, 20	$(8 + 12 + 16 + 20)/4 = 14$ minutes.
B, C, D, A	4, 8, 12, 20	$(4 + 8 + 12 + 20)/4 = 11$ minutes.

Result

Running the shorter jobs first reduces the average turnaround time from 14 minutes to 11 minutes in this example.

8. Why SJF Is Optimal

Turnaround Formula

For four jobs with running times a , b , c , and d , the average turnaround time is:

$$\frac{4a + 3b + 2c + d}{4}$$

Reasoning

The first job contributes to every later completion time, so it has the largest effect on the average. Therefore the shortest job should go first, the next shortest second, and the longest last.

Important Condition

Shortest job first is optimal only when all jobs are available at the same time and their running times are known.

9. SJF Limitation: Arrival Times

Counterexample

Consider five jobs with run times $A = 2$, $B = 4$, $C = 1$, $D = 1$, and $E = 1$. Jobs A and B arrive at time 0. Jobs C, D, and E arrive at time 3.

Order	Why It Happens	Average Turnaround
A, B, C, D, E	SJF initially chooses A because C, D, and E have not arrived yet.	4.6
B, C, D, E, A	Not the immediate SJF choice, but gives a lower average turnaround in this case.	4.4

Lesson

Knowing future arrivals can change the best schedule. Simple SJF only chooses among jobs that are currently available.

10. Shortest Remaining Time Next

SRTN

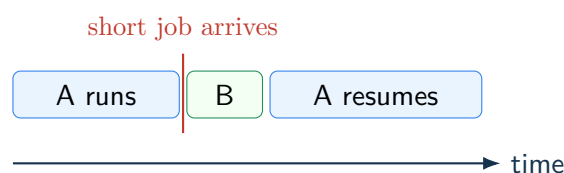
Shortest remaining time next, or **SRTN**, is the preemptive version of shortest job first. The scheduler always runs the process with the smallest remaining time.

How It Works

When a new job arrives, the scheduler compares the new job's total required time with the currently running process's remaining time.

Preemption Rule

If the new job can finish sooner than the current process's remaining time, the current process is suspended and the new job starts.



11. SJF vs. SRTN

Feature	Shortest Job First	Shortest Remaining Time Next
Preemption	Nonpreemptive.	Preemptive.
Choice rule	Run the shortest available job.	Run the job with the shortest remaining time.
New short job arrives	Current job keeps running.	Current job may be suspended.
Main benefit	Minimizes average turnaround when all jobs are available together.	Gives very good service to newly arriving short jobs.
Main assumption	Job lengths are known.	Job lengths and remaining times are known.

12. Big Picture

Main Conclusion

Batch scheduling algorithms try to improve overall job completion behavior. FCFS is simple and fair by arrival order, but can make short I/O-bound jobs wait. SJF reduces average turnaround when job lengths are known. SRTN improves on SJF by preempting long jobs when shorter jobs arrive.

Exam Shortcut

Remember: FCFS is nonpreemptive FIFO; SJF is nonpreemptive shortest known job first; SRTN is preemptive shortest remaining time first.

13. Quick Review

Key Points

- Batch systems run noninteractive jobs.
- Batch scheduling often focuses on throughput, turnaround time, and CPU utilization.
- FCFS runs jobs in arrival order.
- FCFS is simple, fair by queue order, and easy to implement.
- FCFS can make short I/O-bound jobs wait behind long CPU-bound jobs.
- SJF runs the shortest available job first.
- SJF assumes job running times are known in advance.
- SJF minimizes average turnaround time when all jobs are available simultaneously.
- SJF is not always optimal when jobs arrive at different times.
- SRTN is the preemptive version of SJF.
- SRTN runs the job with the least remaining time.
- SRTN can suspend the current process when a shorter job arrives.